7-W1　球壳内的首个异常场

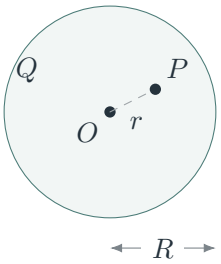
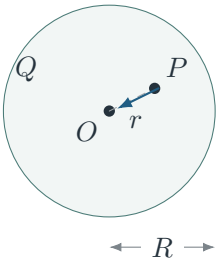


图 7-12　题图　均匀球面与偏离球心的观察点

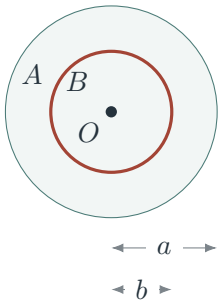


$$\eta > 0 : \quad E_r < 0$$

$$E_r \propto -\eta r$$

图 7-13　解答分析图　内场首项随位移线性变化，正偏离时指向球心

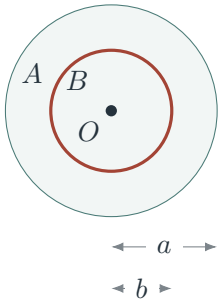
7-W2　电压检测阈值与指数上限



先连通：$V_A = V_B = V$

再断开，外球接地

图 7-14　题图　充电、断开、接地的顺序决定边界约束



q_b 保持，$V_A = 0$

$q'_a = -Cq_b/A$

$V'_b/V = 1 - C/A$

图 7-15　解答分析图　外球保留在原处，并具有非零感应电量

7-W3　感应传递的相对标定

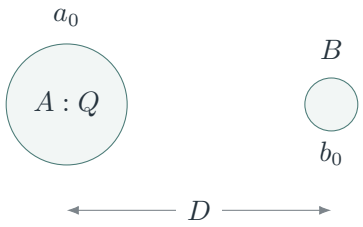
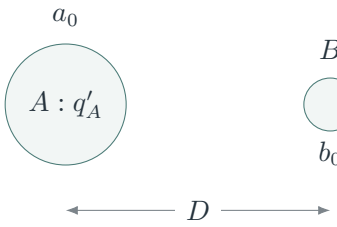


图 7-16　题图　两球远距近似；依次接地并撤地线



$$Q \;\to\; q_B = -b_0 Q/D$$
$$q'_A = a_0 b_0 Q/D^2$$

图 7-17　解答分析图　先保持小球电量，再由大球接地条件求回传电量

7-W4　两类载流子和斜截面

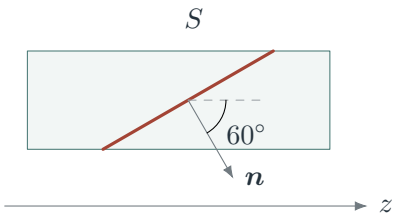
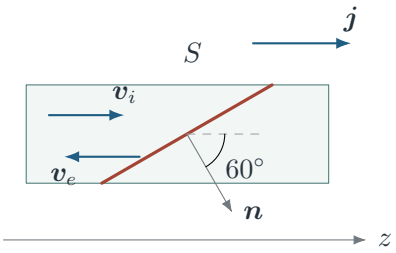


图 7-18　题图　完整斜截面与两类相反漂移的载流子



两种电流贡献同向

$$I = jS_{\perp} = jS\cos\theta$$

图 7-19　解答分析图　负电子电荷使电子漂移与约定电流反向

7-W5　自由体电荷的弛豫

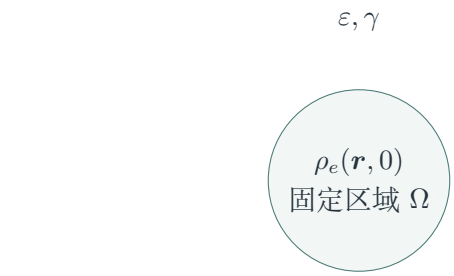


图 7-20　题图　均匀欧姆介质中的固定积分区域

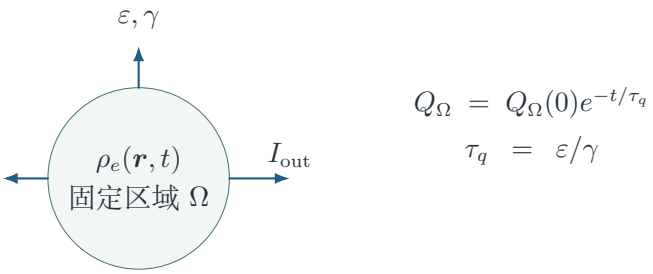


图 7-21　解答分析图　正初始电荷的流出示意；有向通量遵守相同守恒式

7-W6　缓变圆锥导线的电阻和电势

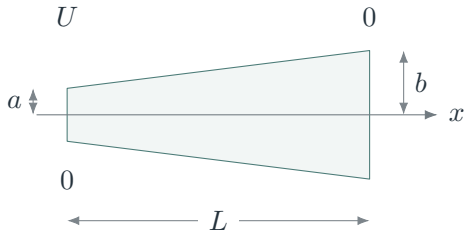


图 7-22　题图　半径线性变化的缓变圆锥台

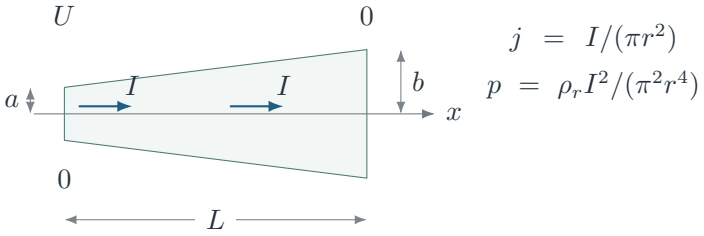


图 7-23　解答分析图　电流相同，窄端电流密度与发热密度更大

7-W7　球形电阻与空间发热

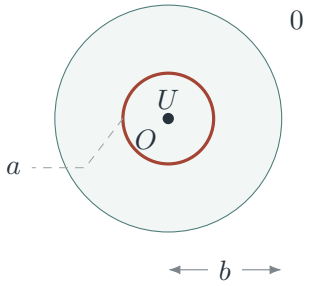
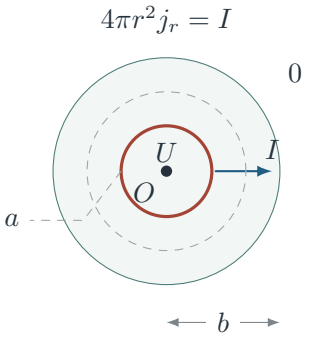


图 7-24　题图　两同心球电极之间均匀导电



$$R = \frac{1/a - 1/b}{4\pi\gamma}$$

$$p(r) \propto r^{-4}$$

图 7-25　解答分析图　同一电流穿过越来越大的球面

7-W8　平行分层导体的有效电导率

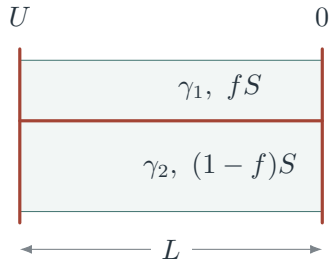


图 7-26　题图　界面平行于电流方向，两条材料等压

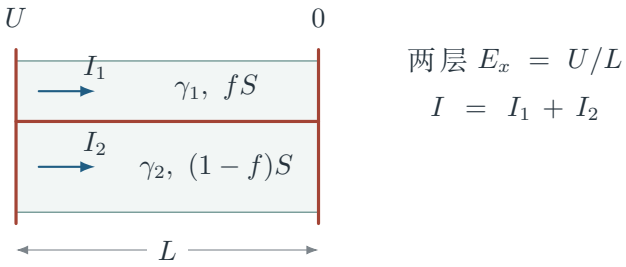


图 7-27　解答分析图　电导率按截面积加权作算术平均

7-W9　电流折射与界面电荷的符号

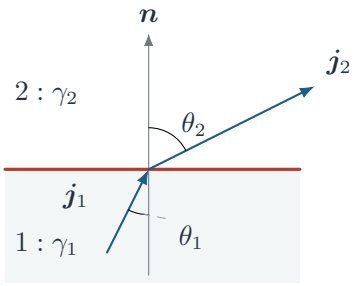
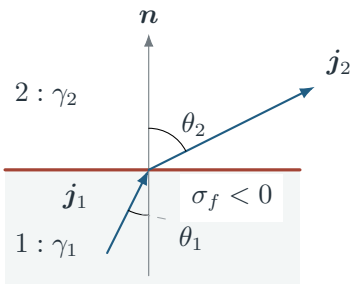


图 7-28　题图　电流方向在两材料界面改变，角从法向起算



$$j_{1n} = j_{2n} = J$$
$$E_{1t} = E_{2t}$$
$$E_{2n} = E_{1n}/4$$

图 7-29　解答分析图　负界面电荷使第二材料法向电场减小

7-W10　自热对伏安关系的影响

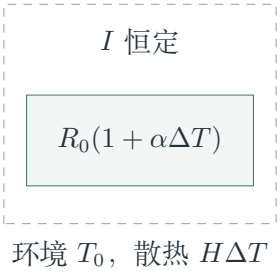


图 7-30　题图　带电阻温度反馈和线性散热的恒流元件

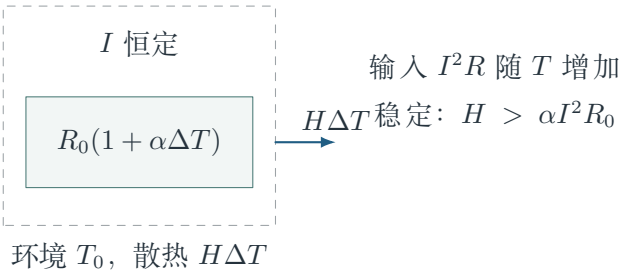


图 7-31　解答分析图　散热斜率大于发热斜率才有稳定温升

7-W11　非线性元件的负载线与微分响应

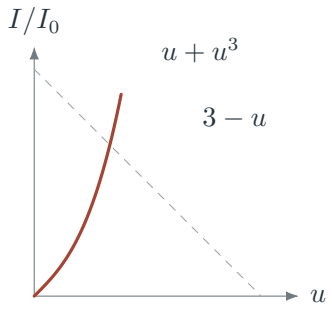
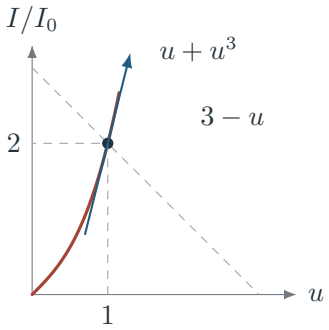


图 7-32　题图　非线性伏安曲线和串联电阻负载线



交点：$u = 1$

$r_d = U_0/(4I_0)$

$\delta U = \delta\mathcal{E}/5$

图 7-33　解答分析图　工作点与切线斜率分别确定直流和小扰动

7-W12　一个电池给另一个电池充电

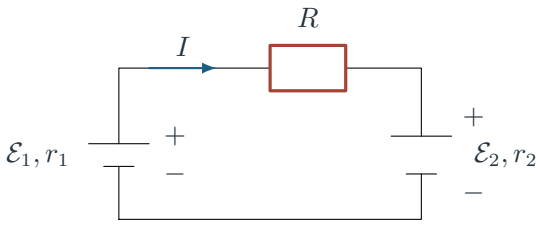


图 7-34　题图　电流进入电池 2 正端，两电动势相反

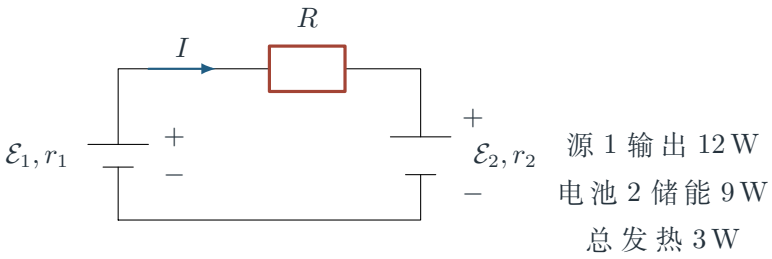


图 7-35　解答分析图　同一回路中区分放电输入、充电储能和内外耗散

7-T1　球壳中心附近的势差

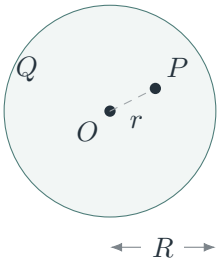
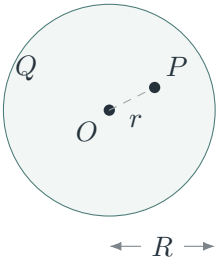


图 7-36　题图　球心与小位移处的电势比较

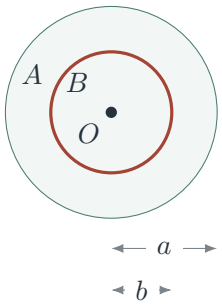


$$\Delta\varphi \simeq kQ\eta/(600R)$$

$$\eta > 0:\ \varphi(P) > \varphi(O)$$

图 7-54　解答分析图　势向外升高对应向心场

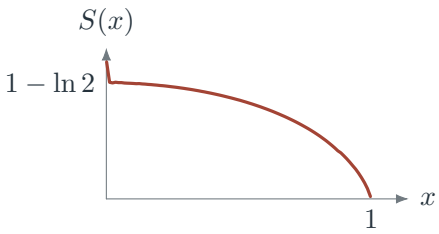
7-T2　几何灵敏度的两个极限



$x = b/a$

比较小内球与窄间隙

图 7-37　题图　固定充电电压下改变内球尺寸

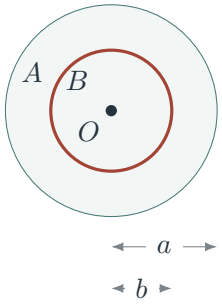


$S(0^+) = 1 - \ln 2$

$S(1^-) = 0$

图 7-55　解答分析图　理想电压灵敏度随内球半径比增大而下降

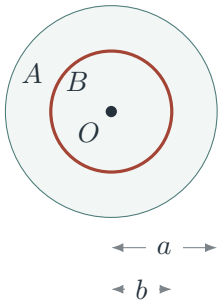
7-T3　校准、零漂和指数上限



校准：$V/500 \ \mapsto \ 80$ 格

零漂阈值：0.020 格

图 7-38　题图　用相对校准把读出格数换成电压



$d/|V| \ = \ 5.00 \times 10^{-7}$

再除以 $S(0.5)$

得到指数上限

图 7-56　解答分析图　检测阈值与几何传递系数分别换算

7-T4　斜截面不是少了一半电流

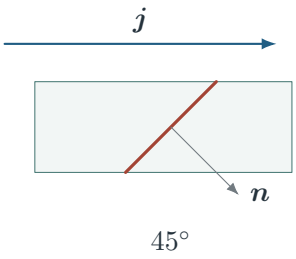
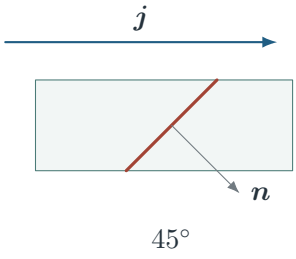


图 7-39　题图　完整斜截面与固定实际面积的小面片



完整截面：$S = \sqrt{2}S_0$

固定面片：$S = S_0$

投影系数均为 $1/\sqrt{2}$

图 7-57　解答分析图　相同角度不意味着相同的积分区域

7-T5　节点电荷积累与非稳恒电流

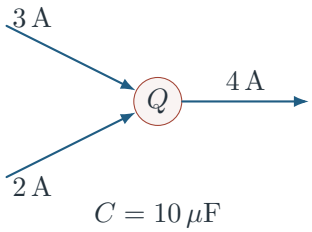


图 7-40　题图　电流不平衡节点与对环境的等效电容

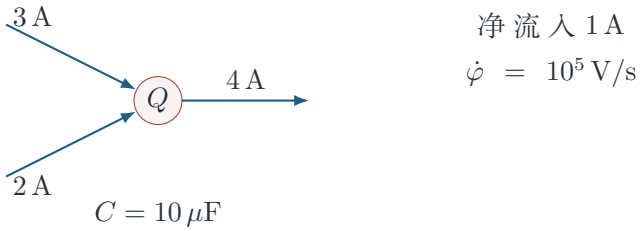


图 7-58　解答分析图　净流入先积累电量，再改变节点电势

7-T6　漂移慢与传输电荷

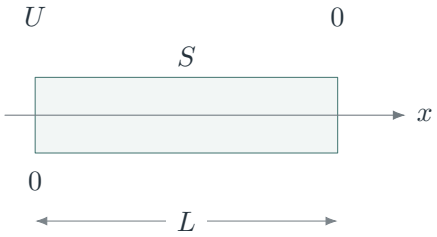


图 7-41　题图　电子漂移与约定电流方向相反

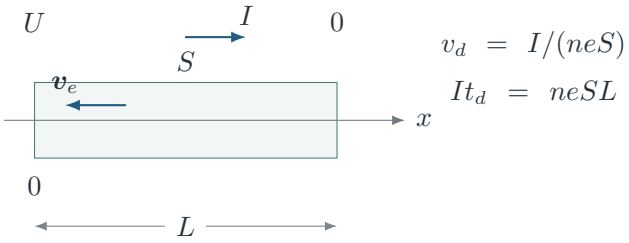


图 7-59　解答分析图　平均漂移穿过全长与电磁信号传播属于不同时间

7-T7　电阻率和截面积同时变化

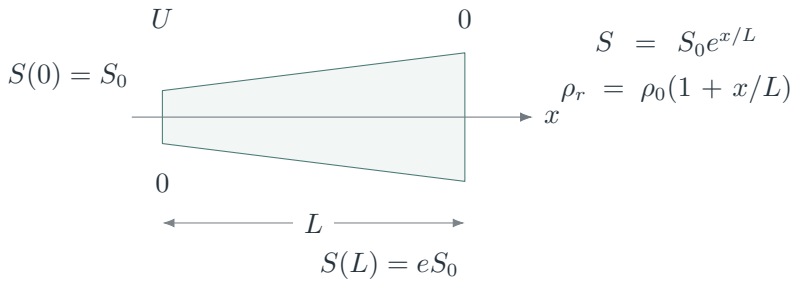


图 7-42　题图　渐变截面积和电阻率都随位置变化

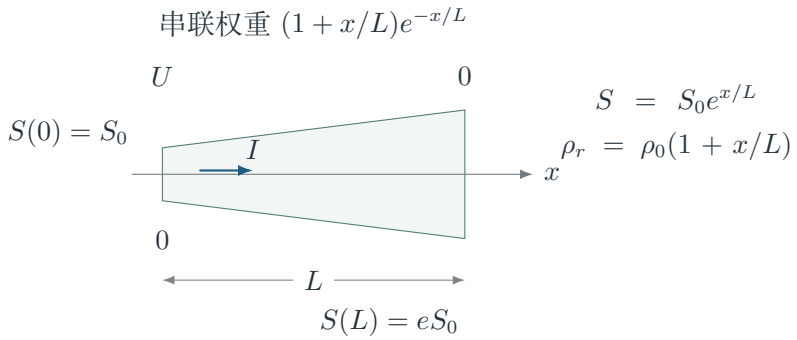


图 7-60　解答分析图　电阻积分与单位长度发热采用同一几何权重

7-T8　两层球形导电介质

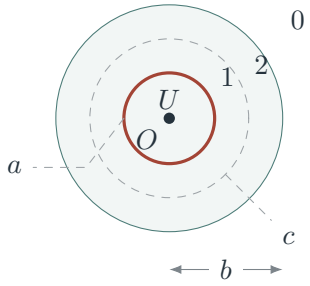
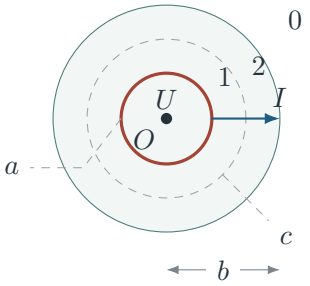


图 7-43　题图　两同心介质层串联，分界半径为 c



$$j_n = I/(4\pi c^2)$$

$$\sigma_f = j_n(\varepsilon_2/\gamma_2 - \varepsilon_1/\gamma_1)$$

图 7-61　解答分析图　同一法向电流通过界面，位移场跳变给面电荷

7-T9　同轴圆柱的径向电阻

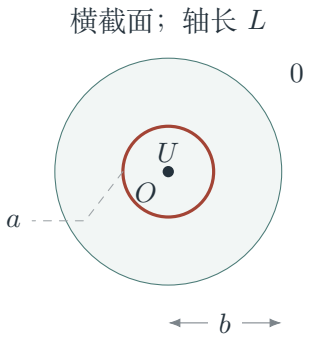
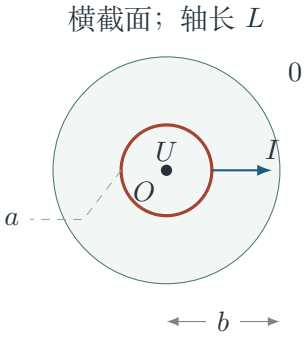


图 7-44　题图　同轴电极之间径向流动，图为横截面



截面 $2\pi rL$

$R = \ln(b/a)/(2\pi\gamma L)$

图 7-62　解答分析图　圆柱面面积随半径一次增长，产生对数电阻

7-T10　串联材料的界面电荷

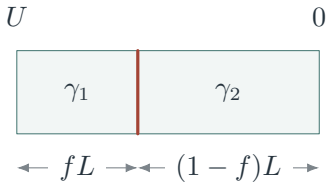
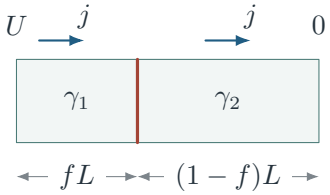


图 7-45　题图　材料界面垂直于电流，两段串联



$$E_i = j/\gamma_i$$

$$\sigma_f = \varepsilon j(1/\gamma_2 - 1/\gamma_1)$$

图 7-63　解答分析图　相同电流密度下，低电导率段具有较大场

7-T11　各向异性：电流与电场可以不平行

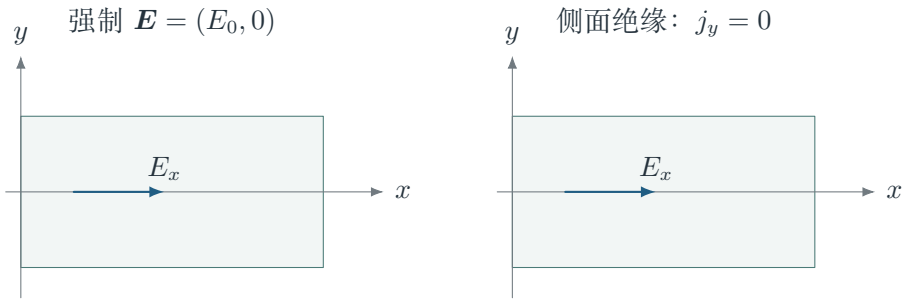


图 7-46　题图　各向异性长条与侧面零法向电流

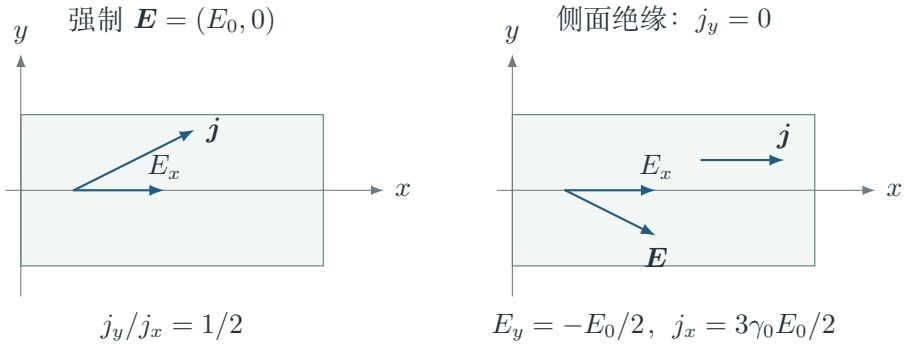


图 7-64　解答分析图　横向场取消横向电流，电场与电流仍不平行

7-T12　接触电阻要单独建模

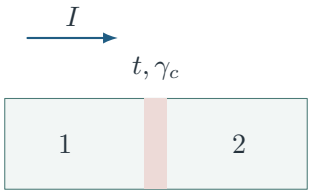
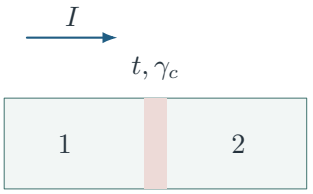


图 7-47　题图　有限厚接触层与两侧串联导体



$$R_c = r_c/S$$
$$U_c = r_c j_n$$
$$P_c = I^2 R_c$$

图 7-65　解答分析图　薄层压缩后保留有限电压降和有限热功率

7-T13　电源内部的场可以逆电流

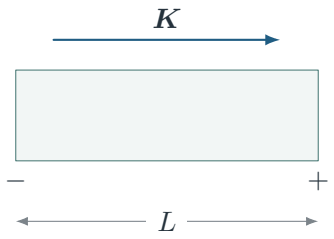
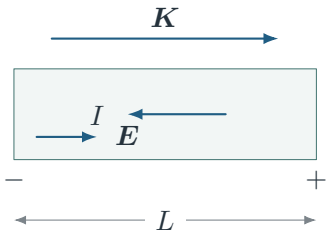


图 7-48　题图　规定电源内部正向从负端到正端



正常放电：$E_x < 0$

$\mathcal{E}I - IU = I^2 r$

图 7-66　解答分析图　非静电力沿电流，静电场逆向，差额成为焦耳热

7-T14　稳恒导电问题的唯一性与最小热耗散

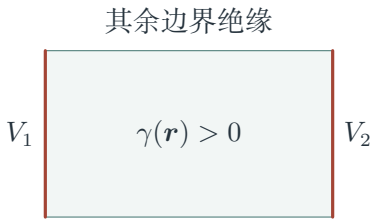


图 7-49　题图　定势电极与绝缘边界组成完整边值问题

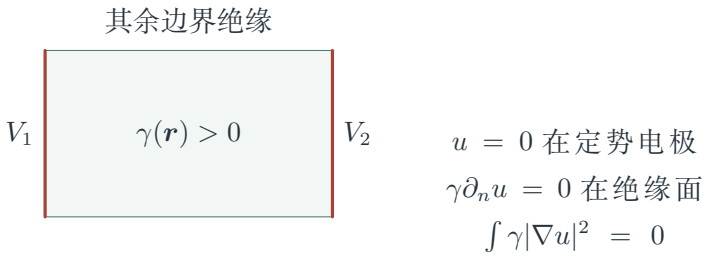


图 7-67　解答分析图　比较同一几何、同一电导率和相同边界电势的试函数

7-T15　绝缘圆孔周围的电流拥挤

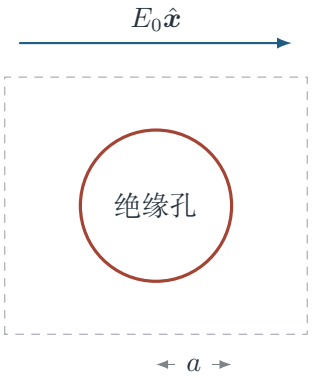
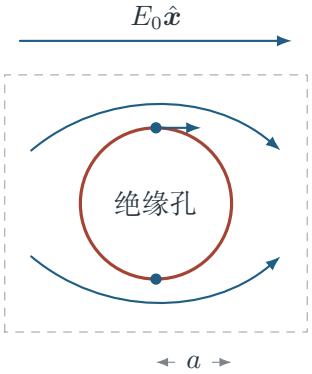


图 7-50　题图　二维无限导电薄片中的圆形绝缘孔



孔边 $j_r = 0$

上下两点：$j_{\max} = 2\gamma E_0$

$p_{\max} = 4p_\infty$

图 7-68　解答分析图　电流在孔上下边缘拥挤，左右端点场为零

7-T16　恒流与恒压的温度稳定性

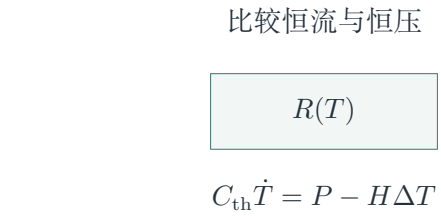


图 7-51　题图　同一电阻元件在两类理想控制条件下

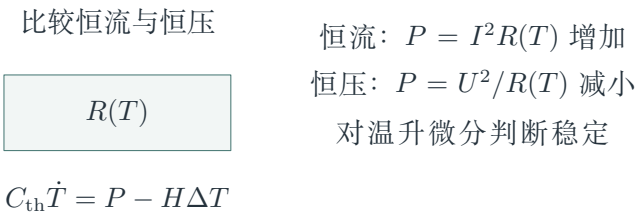
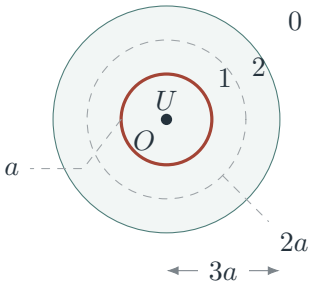


图 7-69　解答分析图　电源约束改变热反馈方向

7-T17　限时综合：两层球形电阻与真实电源

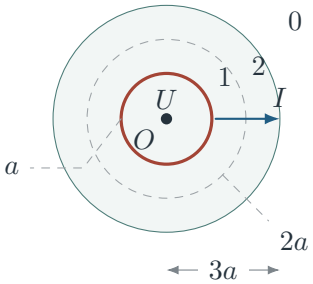


内 层 γ, ε

外 层 $2\gamma, 3\varepsilon$

电 源 $\mathcal{E}, r_0$

图 7-52　题图　两层球介质连接有内阻的电源



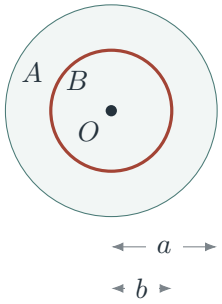
$$R_1 : R_2 = 6 : 1$$

$$\sigma_f > 0$$

$$\mathcal{E}I = I^2 r_0 + I^2 R$$

图 7-70　解答分析图　场比由电导率决定，面电荷还要计介电常数

7-T18　限时综合：反向充电消除零偏

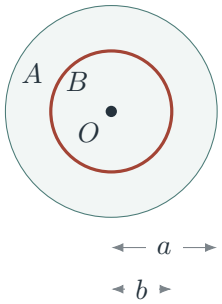


$+V_0 \quad \mapsto \quad v_+$

$-V_0 \quad \mapsto \quad v_-$

两读数共有偏置 v_0

图 7-53　题图　同一几何分别正向和反向充电



差：提取 $\eta S V_0$

和：提取 v_0

奇系统误差仍保留

图 7-71　解答分析图　利用响应奇偶性消除不变偏置，误差界按最坏情况计算